

Module:1 - Database Systems Concepts and Architecture

Need for database systems – Characteristics of Database Approach – Advantages of using DBMS approach - Actors on the Database Management Scene: Database Administrator - Classification of database management systems - Data Models – Schemas and Instances - Three-Schema Architecture - The Database System Environment - Centralized and Client/Server Architectures for DBMSs – Overall Architecture of Database Management Systems.

Unstructured Data

The university has 5600 students. Shaun (ID Number: 160801), 18 years old Communication study. Linh with ID number 160802, majoring in Accounting and is 20 years old. Ahmed from Psychology study program, 19 years old, ID number 160803.

Semi-Structured Data

```
<University>  
  <ID Number="160801">  
    <Name="Shaun">  
    <Age="18">  
    <Program="Communication">  
  <ID Number="160802">  
    <Name="Linh">  
    <Age="20">  
    <Program="Accounting">  
..... </University>
```

Structured Data

ID	Name	Age	Program
160801	Shaun	18	Communication
160802	Linh	20	Accounting
160803	Ahmed	19	Psychology

Difference between Data and Information

Data

- Raw facts of things
- No contextual meaning
- Just numbers and text

Information

- Data with exact meaning
- Processed data and organized context



Data vs. Information

- **Data** are raw facts
- **Information** is the result of processing raw data to reveal meaning
- Information requires *context* to reveal meaning
- Raw data must be *formatted* for storage, processing, and presentation
- Data are the foundation of information, which is the bedrock of **knowledge**



INTRODUCTION TO DATABASES

A *database* is a collection of data organized to serve many application. By using centralized data it can easily be accessed, managed, and updated.

And a *DATABASE MANAGEMENT SYSTEM* (DBMS) is a collection of programs that enables you to store, modify, and extract information from a database.

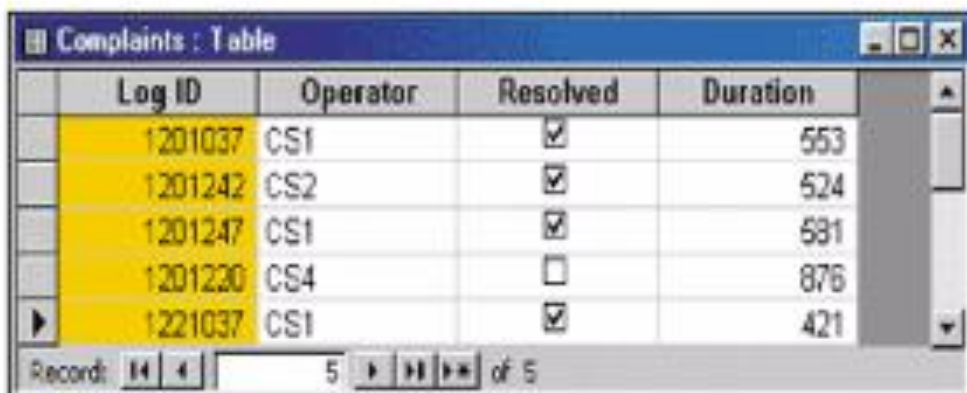
Introducing the Database and the DBMS

- Database: shared, integrated computer structure that stores a collection of data
 - End-user data: raw facts of interest to the end user
 - **Metadata**: data about data
- Metadata provides description of data characteristics and relationships in data
 - Complements and expands value of data
- **Database management system (DBMS)**: collection of programs
 - Manage structure and control access to data

ELEMENT OF DATABASE



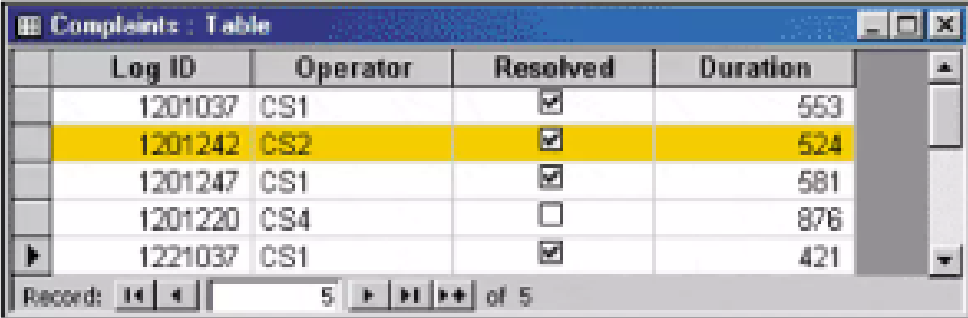
FIELDS: A field is part of a record and contains a single piece of data for the subject of the record. In the database table illustrated in Figure 4, each record contains four fields:



Log ID	Operator	Resolved	Duration
1201037	CS1	☒	553
1201242	CS2	☒	524
1201247	CS1	☒	581
1201230	CS4	☐	876
1221037	CS1	☒	421

Record: 14 5 of 5

RECORDS: Data is stored in records. A record is composed of fields and contains all the data about one particular person, company, or item in a database. In this database, a record contains the data for one customer support incident report. Records appear as rows in the database table. A record for Log ID 1201242 is highlighted in Below mentioned Figure.



Log ID	Operator	Resolved	Duration
1201037	CS1	☒	553
1201242	CS2	☒	524
1201247	CS1	☒	581
1201220	CS4	☐	876
1221037	CS1	☒	421

TABLES: A database table is composed of records and fields that hold data. Tables are also called datasheets. Each table in a database holds data about a different, but related, subject.



Log ID	Operator	Resolved	Duration
1201037	CS1	☒	553
1201242	CS2	☒	524
1201247	CS1	☒	581
1201220	CS4	☐	876
1221037	CS1	☒	421

Record: 1 4 5 of 5

Role and Advantages of the DBMS



- DBMS is the intermediary between the user and the database
- Database structure stored as file collection
- Access database through the DBMS
- DBMS enables data to be shared
- DBMS integrates many users' views of the data

Types of Databases



- Databases can be classified according to:
 - Number of users
 - Database location(s)
 - Expected type and extent of use
- **Single-user database** supports only one user at a time
 - **Desktop database** – single-user, runs on PC
- **Multiuser database** supports multiple users at the same time
 - **Workgroup database** supports a small number
 - **Enterprise database** supports a large number

Types of Databases (continued)



- **Centralized database:** data located at a single site
- **Distributed database:** data distributed across several different sites
- **Operational database:** supports a company's day-to-day operations
 - **Transactional or production database**
- **Data warehouse:** stores data used for tactical or strategic decisions



Types of Databases (continued)

- **Unstructured data** exist in their original state
- **Structured data** result from formatting
 - Structure applied based on type of processing to be performed
- **Semistructured data** have been processed to some extent
- **Extensible Markup Language (XML)** represents data elements in textual format
- **XML database** supports semistructured XML data

Why Database Design is Important

- **Database design** focuses on design of database structure used for end-user data
 - Designer must identify database's expected use
- Well-designed database:
 - Facilitates data management
 - Generates accurate and valuable information
- Poorly designed database:
 - Causes difficult-to-trace errors



ADVANTAGES OF USING A DATABASE APPROACH

- **Flexible Data Access.**
- **Improved Data Integrity (purity,).**
- **Improved Data Security.**
- **Data Independence.**
- **Reduced Data Redundancy (excess).**
- **Ability to Share and Relate Data**
- **Standardisation of Data.**
- **Increased Productivity.**



University Database Example

- In this text we will be using a university database to illustrate all the concepts
- Data consists of information about:
 - Students
 - Instructors
 - Classes
- Application program examples:
 - Add new students, instructors, and courses
 - Register students for courses, and generate class rosters
 - Assign grades to students, compute grade point averages (GPA) and generate transcripts



View of Data

- A database system is a collection of interrelated data and a set of programs that allow users to access and modify these data.
- A major purpose of a database system is to provide users with an abstract view of the data.
 - Data models
 - A collection of conceptual tools for describing data, data relationships, data semantics, and consistency constraints.
 - Data abstraction
 - Hide the complexity of data structures to represent data in the database from users through several levels of data abstraction.



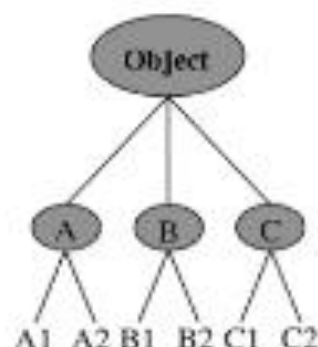
Data Models

- A collection of tools for describing
 - Data
 - Data relationships
 - Data semantics
 - Data constraints
- Relational model
- Entity-Relationship data model (mainly for database design)
- Object-based data models (Object-oriented and Object-relational)
- Semi-structured data model (XML)
- Other older models:
 - Network model
 - Hierarchical model

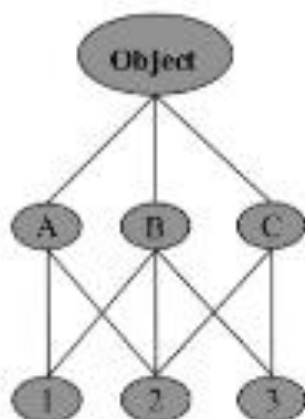


DATA MODELS

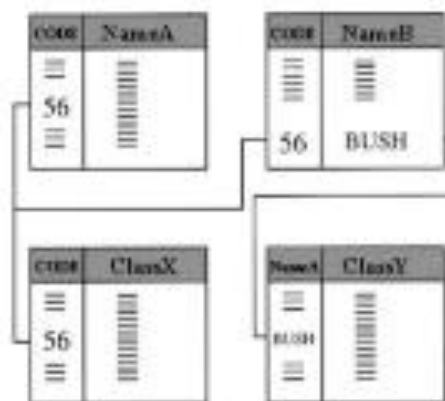
1. HIERARCHICAL
2. NETWORK
3. RELATIONAL
4. OBJECT



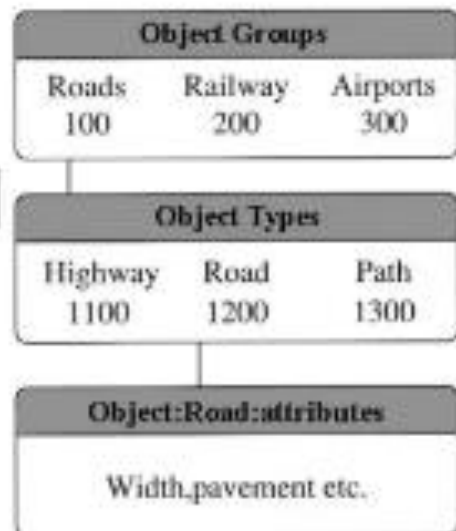
(a) Hierarchical model



(b) Network model



(c) Relation of model

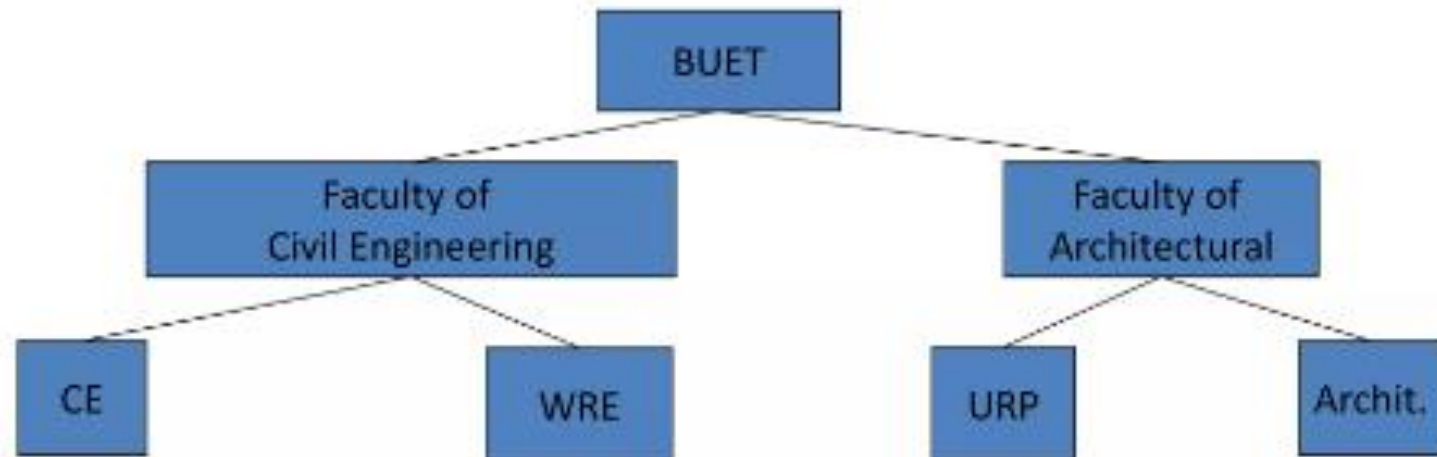


(d) Object oriented model

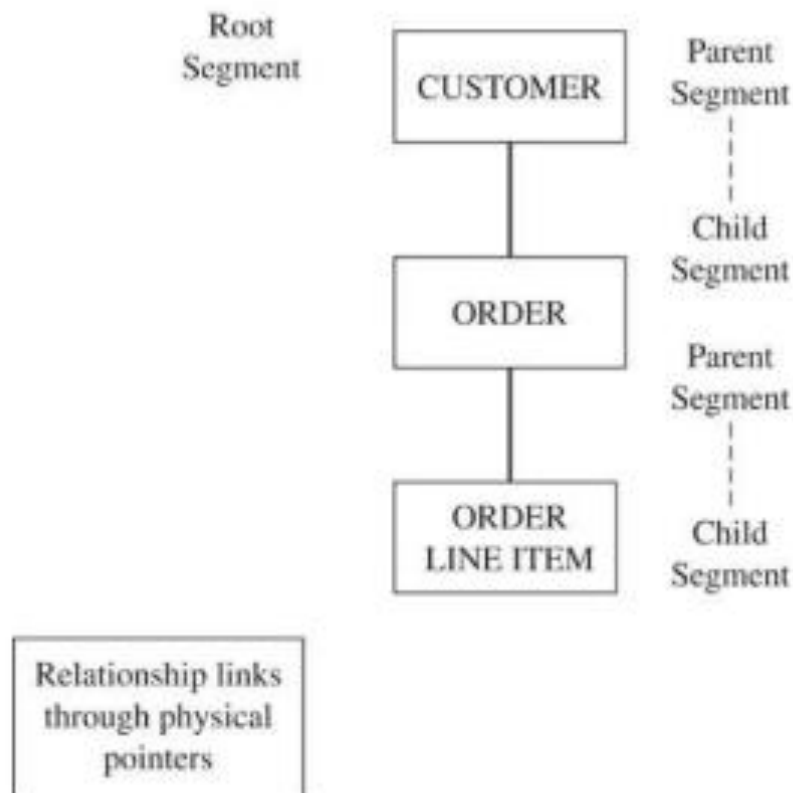


1. HIERARCHICAL MODEL

Stores data as hierarchically related to each other. Record shape are tree structure.

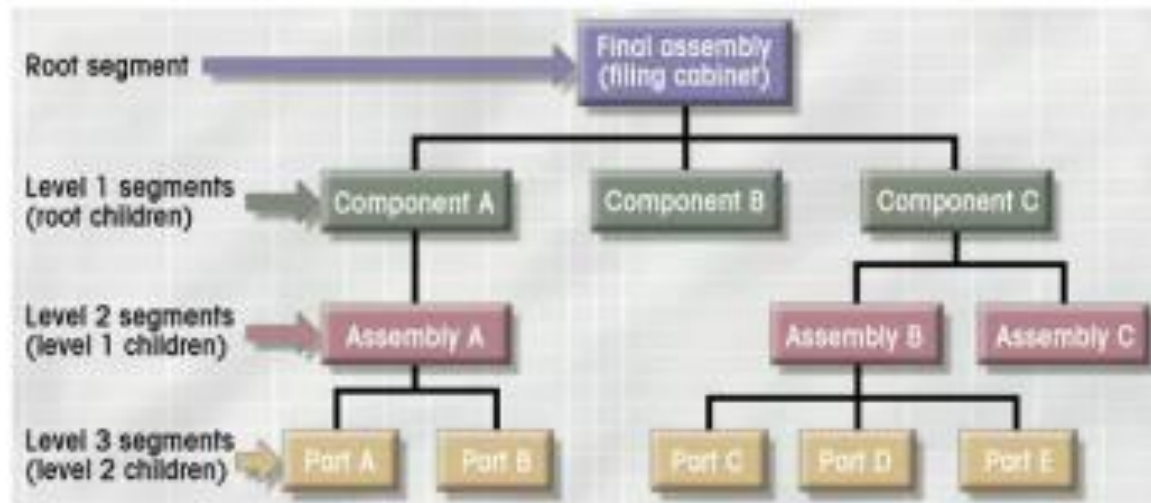


HIERARCHICAL DATABASE MODEL



HIERARCHICAL DATABASE MODEL

- Logically represented by an upside down tree
 - Each parent can have many children
 - Each child has only one parent



HIERARCHICAL MODEL

- Several records or files are hierarchically related with each other. For example, an organization has several departments, each of which has attributes such as name of director, number of staffs, annual products etc.
- Each department has several divisions with attributes of name of manager, number of staffs, annual products etc.
- Then each division has several sections with attributes such as name of head, number of staff, number of PCs etc.

Advantage and Disadvantages of Hierarchical Model



■ Advantages

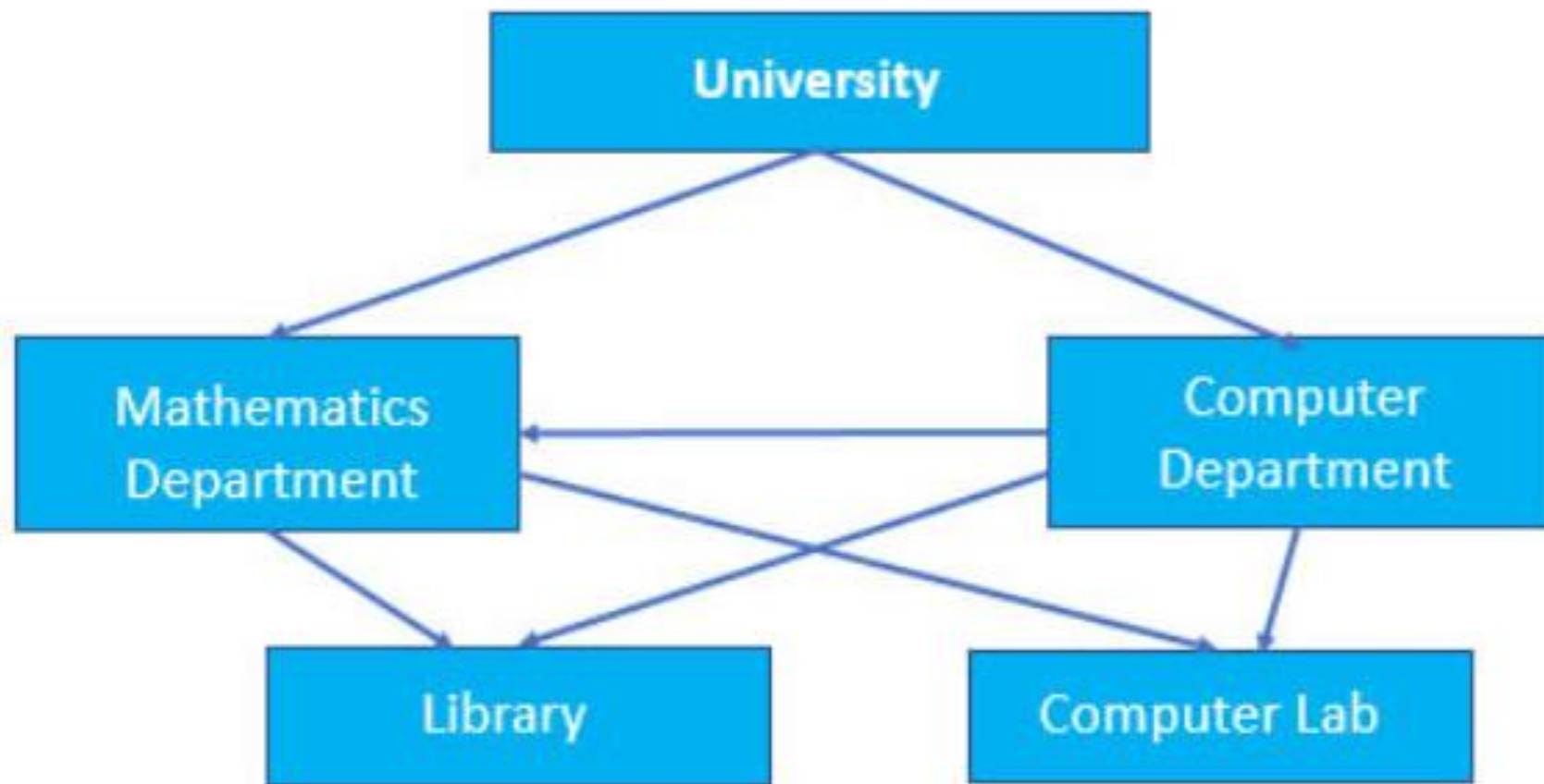
- ☐ High speed access to large databases
- ☐ Easy to update- (to add or delete new nodes)

■ Disadvantages

- ☐ Links are only possible in Vertical Direction (from top to bottom) but not for horizontal or diagonal unless they have same parents.
- ☐ For example, it is hard to find what is the relation between URP and DCE from this data model.

2. NETWORK DATABASE MODEL

- Doesn't force data into hierarchical levels
- Owner/Member relationships:
 - Owner record type
 - Member record type
- Each owner may have one or more member types
- Each member type and corresponding owner record type form set, which represents relationship



The major advantage of network model are-

- 1.) Conceptual simplicity
- 2.) Capability to handle more relationship
- 3.) Ease of data
- 4.) Data integrity
- 5.) Data independence
- 6.) Database standards

Disadvantages

Operational difficulties

it is complex and difficult to design, implement, and maintain

Historical Roots: Files and File Systems

- Reasons for studying file systems:
 - Complexity of database design easier to understand
 - Understanding file system problems helps to avoid problems with DBMS systems
 - Knowledge of file system useful for converting file system to database system
- **File systems** typically composed of **collection of file folders, each tagged and kept in cabinet**
 - Organized by expected use



Purpose of Database Systems

In the early days, database applications were built directly on top of file systems, which leads to:

- Data redundancy and inconsistency: data is stored in multiple file formats resulting in duplication of information in different files
- Difficulty in accessing data
 - Need to write a new program to carry out each new task
- Data isolation
 - Multiple files and formats
- Integrity problems
 - Integrity constraints (e.g., account balance > 0) become “buried” in program code rather than being stated explicitly
 - Hard to add new constraints or change existing ones



Purpose of Database Systems (Cont.)

- Atomicity of updates
 - Failures may leave database in an inconsistent state with partial updates carried out
 - Example: Transfer of funds from one account to another should either complete or not happen at all
- Concurrent access by multiple users
 - Concurrent access needed for performance
 - Uncontrolled concurrent accesses can lead to inconsistencies
 - Ex: Two people reading a balance (say 100) and updating it by withdrawing money (say 50 each) at the same time
- Security problems
 - Hard to provide user access to some, but not all, data

Database systems offer solutions to all the above problems



3. RELATIONAL MODEL

- Based on two important concepts:
 - **Key of relation** - one to one, one to many, many to many
 - **Primary attribute** – which can't be duplicate

Student Table

Student ID	Name	CourseID
1	Mr. X	001
2	Mr. X	002
3	Mr. Y	003

Course table

Cour seID	Title	Cre dit
001	RS & GIS in WM	3
002	Watershed Hydrology	3
003	Risk Management	3



Many to many relationship



WHAT IS A RELATIONAL DATABASE?

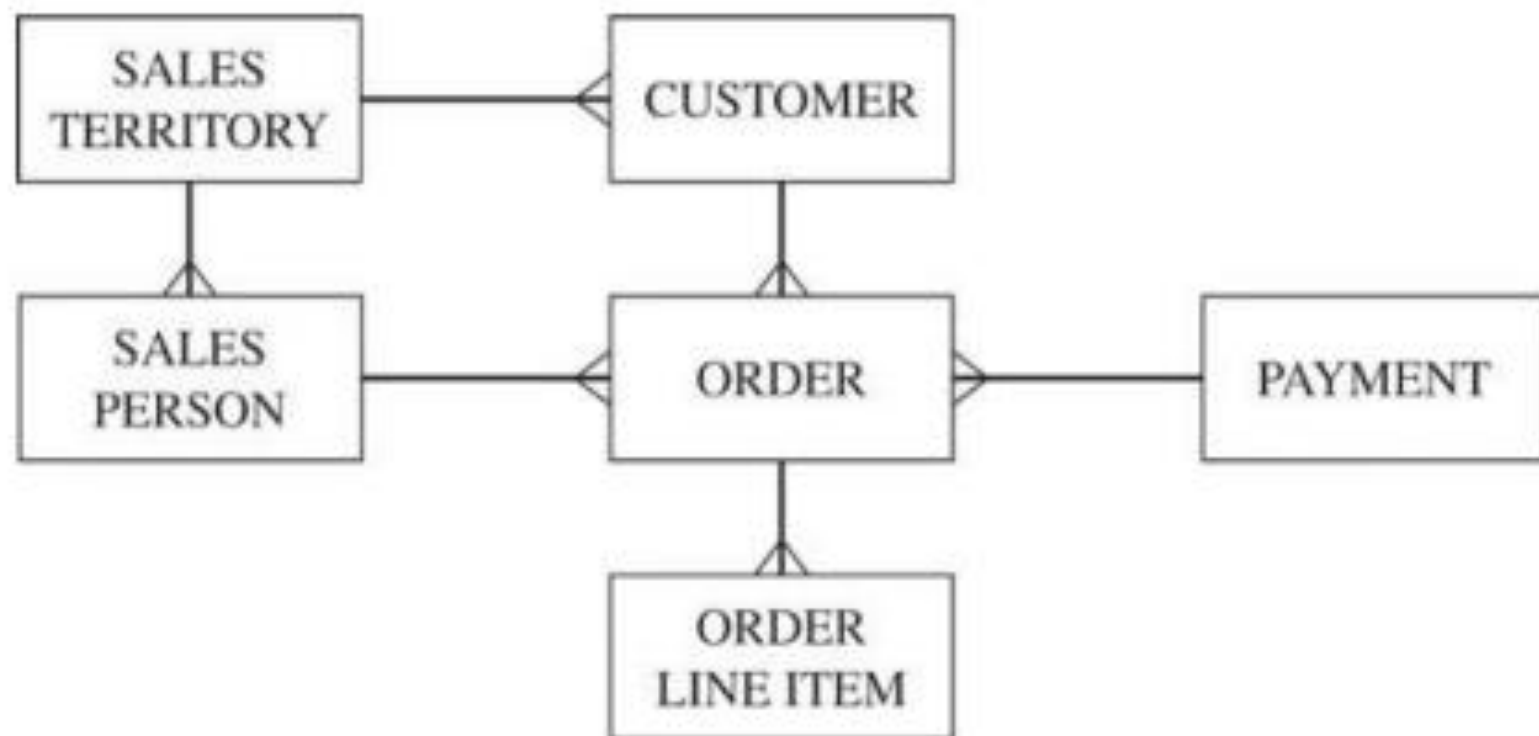
- A database is more than just a collection of information. Such as student and course information, faculty and grades.
- A database is a representation of the people and things your business needs to operate, and the way those people and things relate to each other.
- A database system supports the business rules defined by the customer.



RELATIONSHIP TYPES

1. **One-to-One** : relationship is single valued in both directions. A manager manages one department; a department has only one manager.
2. **One-to-Many** : relationship is multi-valued in one direction - one row in the parent table is associated with many rows in the dependent table. One department has many employees.
3. **Many-to-Many** : relationships are multi-valued in both directions. This type of relationship can be expressed in a table with a column for each entity. (crosswalk table) - An employee can work on more than one project, and a project can have more than one employee assigned. Employee, Project, and Employee/Project tables.

Relational Database Model



Relationship links NOT through physical pointers, but by foreign keys



WHAT IS QUERY LANGUAGE?

- Query language (QL) refers to any computer programming language that requests and retrieves data from database and information systems by sending queries.



WHAT IS QUERY ?

A query is a request for information from a database.

SQL



What is it?

Structured Query Language

- Used in ORACLE and other DB systems
- Non-procedural - i.e. Specify what you want not how to get it
- SQL - (also pronounced SEQUEL)
directly related to the development of the
RELATIONAL MODEL by E.F.Codd.



SQL

- SQL is used to perform query in relations databases.
- For example, find the name of the student who took more than or equal to 6 credit hour in this term

```
SELECT Student.Name, Course.Credit  
FROM Student, Course  
WHERE Student.CourseID = Course.CourseID  
AND Credit >= 6
```

- The answer is :
Mr. X 6



Advantage of Relational Database

- Advantages

- there is no redundancy.
- type of building of an owner can be changed without destroying the relation between type and rate.
- a new type of building for example "Clay" can be inserted. (row insert is easy).

- Disadvantages

- Require a number of tables and relationship
- Its difficult to add a new column in the table.



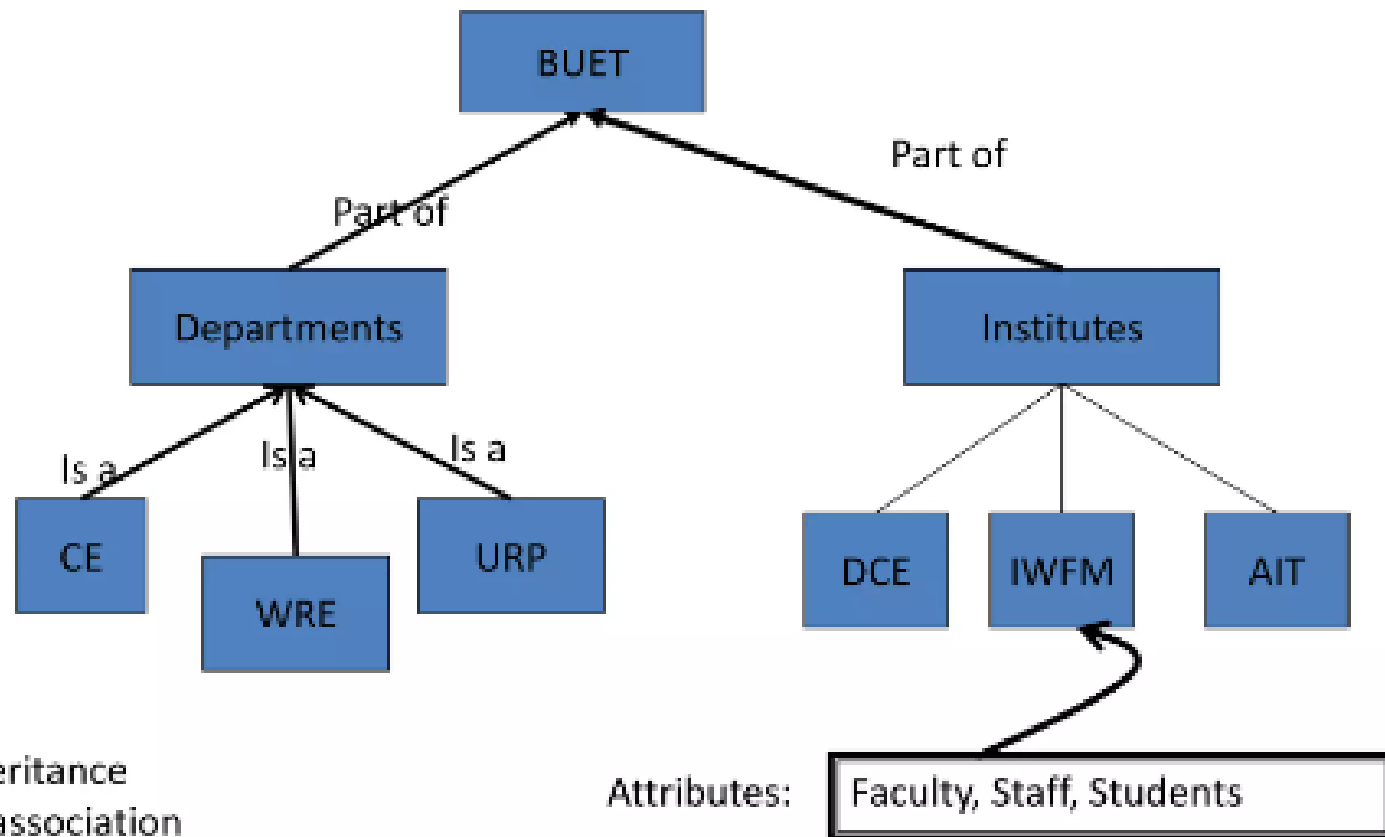
4. OBJECT DATABASES

- Current generation systems have a need to handle complex data for complex applications such as
 - computer aided design
 - computer aided software engineering
 - geographic information systems
 - interactive web sites
- Relational systems are inadequate for these systems
 - Why do you think this is?

- Object-oriented
 - extend a programming language such as Java with persistency and a query language
- Object-relational
 - extend a current RDBMS (e.g. Oracle) with object-oriented extensions



Object Oriented Model





Database Applications Examples

- Enterprise Information
 - Sales: customers, products, purchases
 - Accounting: payments, receipts, assets
 - Human Resources: Information about employees, salaries, payroll taxes.
- Manufacturing: management of production, inventory, orders, supply chain.
- Banking and finance
 - customer information, accounts, loans, and banking transactions.
 - Credit card transactions
 - Finance: sales and purchases of financial instruments (e.g., stocks and bonds; storing real-time market data
- Universities: registration, grades



Database Applications Examples (Cont.)

- Airlines: reservations, schedules
- Telecommunication: records of calls, texts, and data usage, generating monthly bills, maintaining balances on prepaid calling cards
- Web-based services
 - Online retailers: order tracking, customized recommendations
 - Online advertisements
- Document databases
- Navigation systems: For maintaining the locations of various places of interest along with the exact routes of roads, train systems, buses, etc.



Levels of Abstraction

- **Physical level:** describes how a record (e.g., instructor) is stored.
- **Logical level:** describes data stored in database, and the relationships among the data.

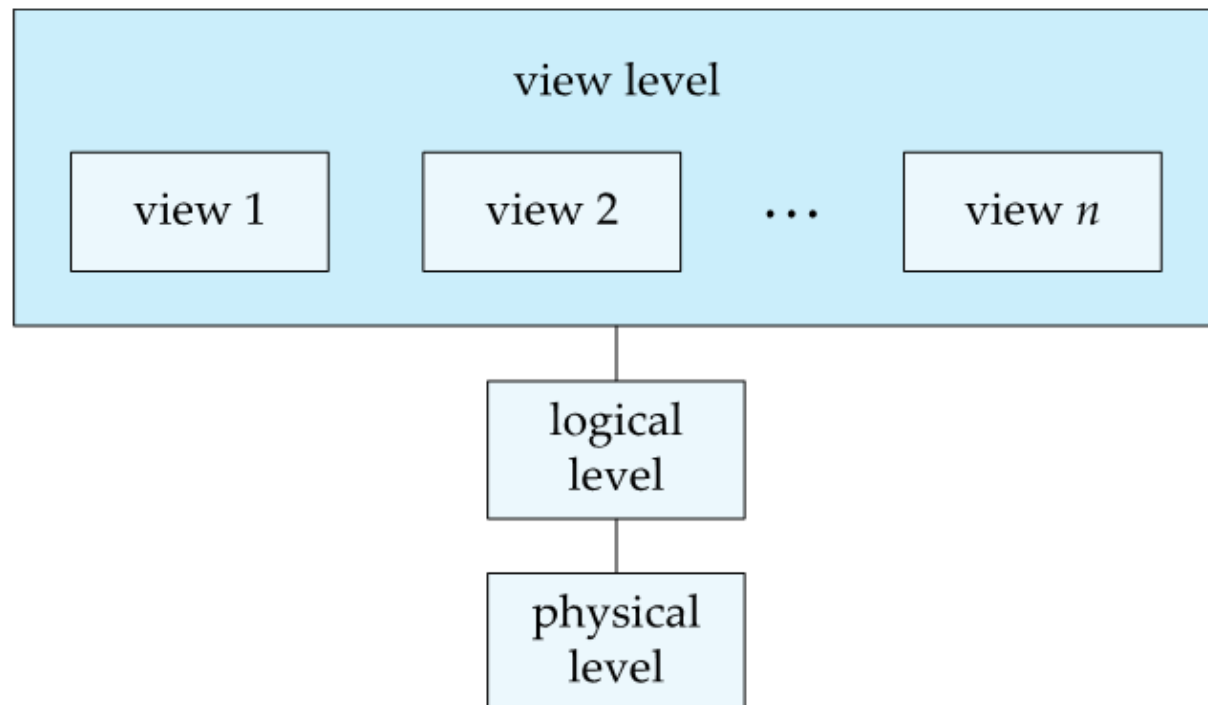
type *instructor* = **record**

```
ID : string;  
name : string;  
dept_name : string;  
salary : integer;  
end;
```

- **View level:** application programs hide details of data types. Views can also hide information (such as an employee's salary) for security purposes.

View of Data

An architecture for a database system





Actors on the scene

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○ Database administrators (DBA):

Responsible for **authorizing access** to the database, for co-coordinating and monitoring its use, acquiring software, and hardware resources, controlling its use and monitoring efficiency of operations.

Functions of DBA:

- Defining the conceptual schema.
- Defining the physical schema.
- Defining the security and Integrity constraints.
- Defining Backup and Recovery procedures.
- Storage structures and access methods definition.
- Granting of authorization for data access.

Categories of End-users



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Different types of end users:

1) Casual: Access database occasionally when needed

2) Naïve or Parametric: They make up a large section of the end-user population.

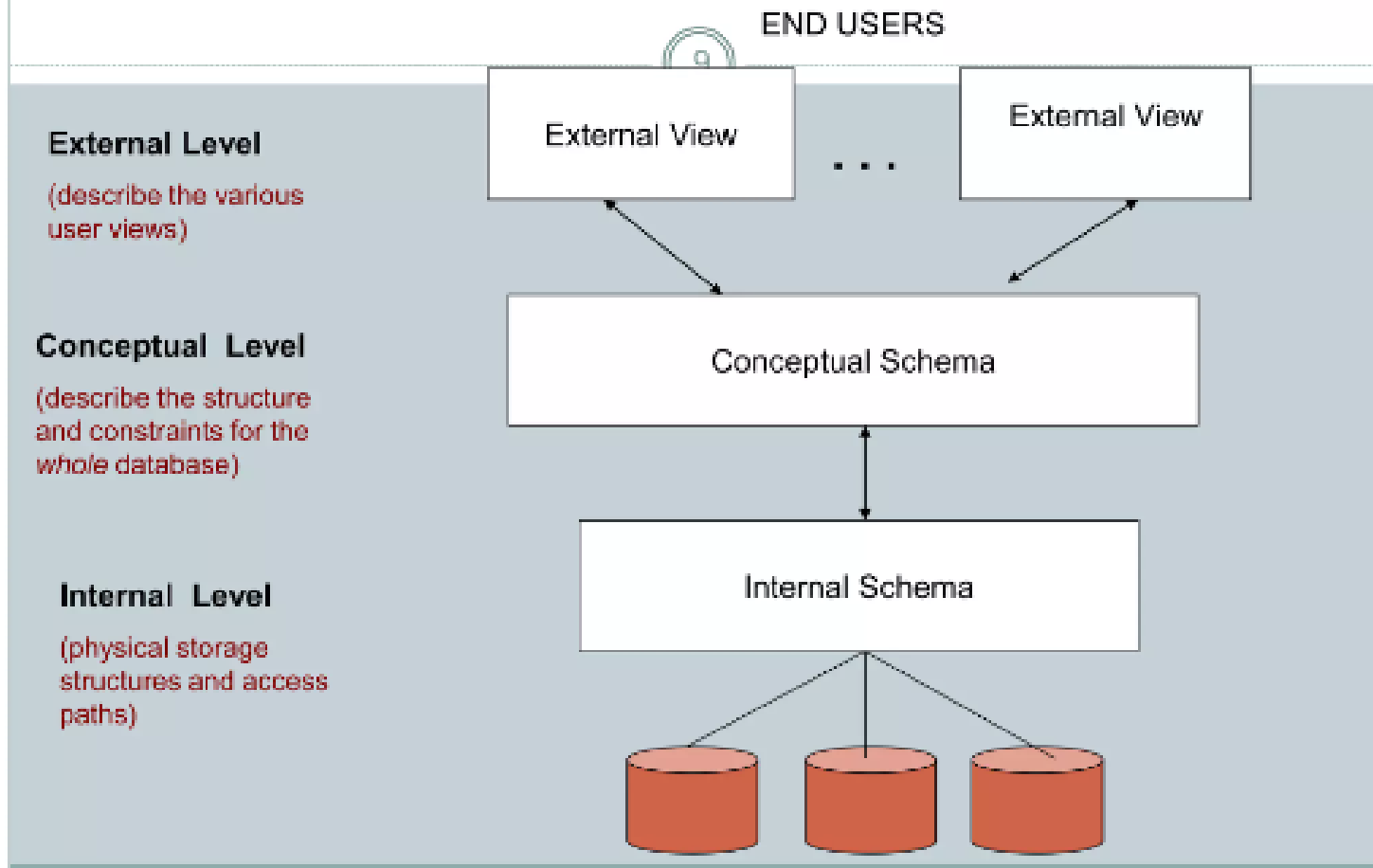
They use previously well-defined **functions** in the form of “canned transactions” (using **Queries and Updates**) against the database.

OR

Users who interact with the system by using the application programs that have previously written, they are **unsophisticated users**.

Examples A bank-tellers or reservation clerks who do this activity for an entire shift of operations.

Three-Schema Architecture



Three-Schema Architecture

Example:



External View

Customer Name
Customer Address
Customer SSN

Logical Record 1

10

Customer Name
Customer Address
Customer SSN

Logical Record n

Conceptual View

Customer Name: String
Customer Address: String
Customer SSN: Number (PK)

Internal View

Name: String length 25
Address: String length 40
SSN: Number length 10

Data Independence

It is the capacity to **change the schema** at **one level database system without** having to **change the schema at the next level**.

Two types of Data independence:

- Physical data independence
- Logical data independence

1. Physical Data Independence :

The physical data independence is basically used to separate conceptual levels from the internal/physical levels. It is easy to achieve physical data independence. With this type of independence, **user is able to change the physical storage structures** should not affect on the conceptual schema.

Examples of changes under Physical Data Independence :

It is by the use of new storage device like Hard Drive or Magnetic Tapes

Modifying the file organization technique in the Database

Switching to different data structures.

Changing the access method.

Modifying indexes.

To change the compression techniques or hashing algorithms.

To change the Location of Database from say C drive to D Drive.

2. Logical Data Independence :

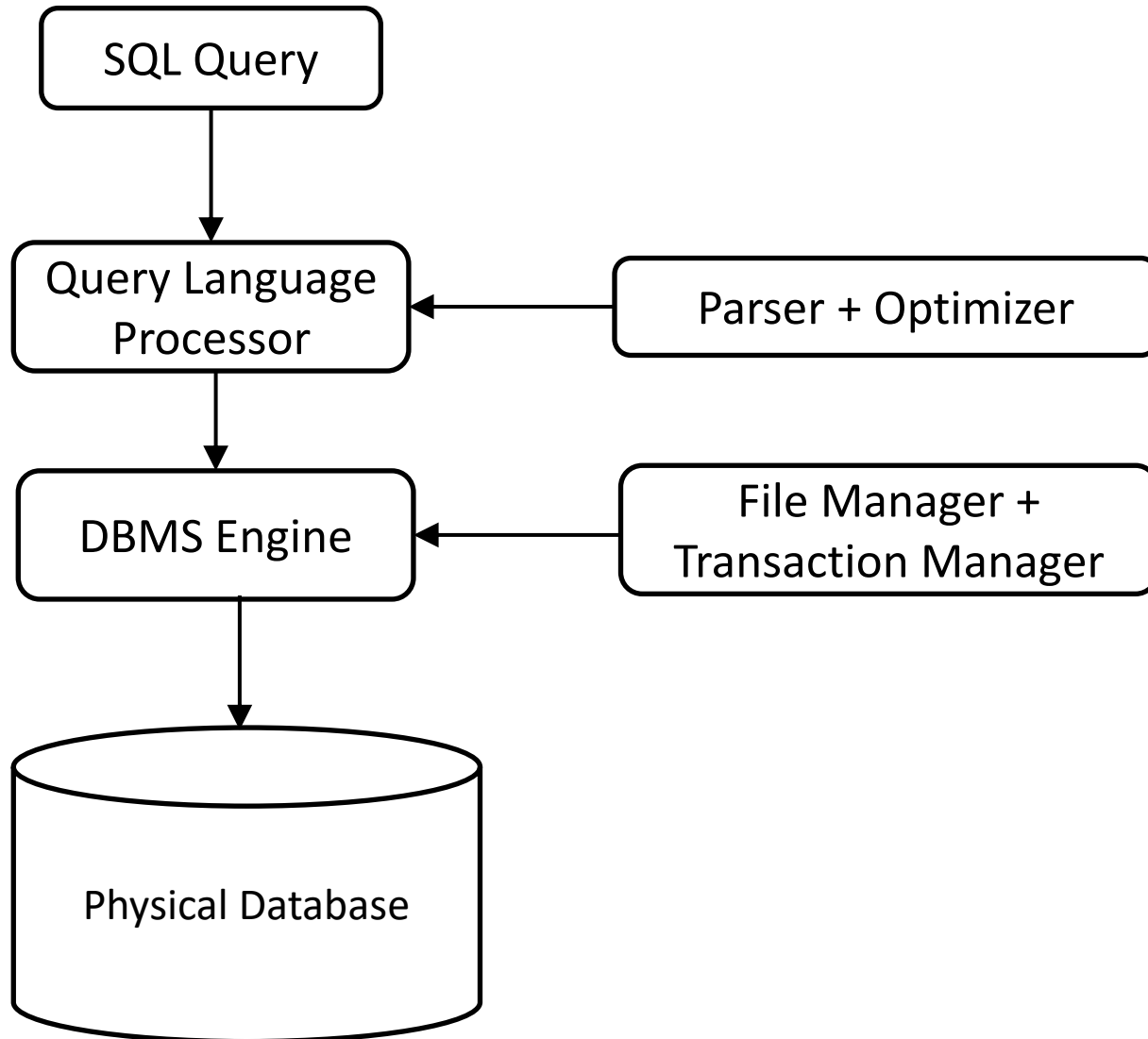
Logical Data Independence is used to change the conceptual scheme without changing the following things :

- External views
- External API or programs

Examples of changes under Logical Data Independence

- To Add/Modify/Delete a new attribute, entity or relationship is possible without a rewrite of existing application programs
- Merging two records into one
- To break an existing record i.e to divide the record into two or more records

SQL Architecture



SQL COMMANDS

DDL – Data Definition Language
DQL – Data Query Language
DML – Data Manipulation Language
DCL – Data Control Language
TCL – Transaction Control Language

DDL

CREATE

DROP

ALTER

TRUNCATE

RENAME

DML

INSERT

UPDATE

DELETE

CALL

LOCK

TCL

COMMIT

SAVEPOINT

ROLLBACK

SET
Transaction

SET
Constraint

DQL

SELECT

DCL

GRANT

REVOKE

- When you are executing an SQL command for any RDBMS, the system determines the best way to carry out your request and SQL engine figure out how to interpret the task.
- There are various components included in the process. These components are Query Dispatcher, Optimization Engines, Classic Query Engine and SQL query engine etc..
- Classic query engine handles all non-SQL queries but SQL query engine won't handle logical files. These are the components of SQL: MySQL, MS SQL server, Oracle, MS Access.



SQL Commands

- The standard SQL commands to interact with relational databases are CREATE, SELECT, INSERT, UPDATE, DELETE and DROP. These commands can be grouped based on their nature.

DDL-Data Definition Language

- DDL defines the conceptual schema providing a link between the logical(the way the user views the data) and the physical (the way in which the data is stored physically) structures of the database.
- The logical structure of a database is called a schema. A subschema is the way a specific application views the data from the database.

Functions of DDL

- Defines the physical characteristics of each record, filled in the record, field's data type, field's length. Field's logical name and also specify relationship among those records.
- Describes schema and subschema.
- Indicates the keys of records.
- Provides means for associating related records or fields.
- Provides data security measures.
- Provides for the logical and physical data independence.;

- Few of the basic commands of DDL are:

Command	Description
CREATE	Creates a new table, a view of table or other object in database.
ALTER	Modifies an existing database object, such as a table.
DROP	Deletes an entire table, a view of table or other object in the database

DML-Data Manipulation Language

- Provides the data manipulation techniques like selection, insertion, deletion, update, modification, replacement, retrieval, sorting and display of data or records.
- Facilitates the use of relationship between the records.
- Enables the user and application program to be independent of the physical data structures and database structures maintenance by allowing to process data on a logical or symbolic basis.
- Provides for independence of programming languages by supporting several high level programming languages like COBOL, PL/1 and C++.

- Few DML commands are:

Command	Description
INSERT	Creates a record
UPDATE	Modifies a record
DELETE	Deletes a record

DCL-Data Control Language

Command	Description
GRANT	Gives a privilege to user
REVOKE	Takes back privileges granted from users

DQL-Data Query Language

Command	Description
SELECT	Retrieves certain records from one or more tables.

Data types in SQL

- SQL data type is an attribute that specifies type of data of any object.
- Each column, variable and expression has related data type in SQL.
- SQL servers offers six categories of data types for your use:

EXACT NUMERIC DATA TYPES

DATA TYPE	FROM	TO
int	-2,147,483,648	2,147,483,647
numeric	$-10^{38}+1$	$10^{38}-1$

Floating point numeric data types:

DATA TYPE	FROM	TO
Float	-179E+308	1.79E+308
Real	-3.40E+38	3.40E+38

Date and Time data types:

DATA TYPE	FROM	TO
datetime	Jan 1, 1753	Dec 31, 9999

Date	Stores a date like MARCH 26, 2014
Time	Stores a time of day like 12:30PM

Note: Here, date time has 3.33 milliseconds accuracy whereas small datetime has 1 minute accuracy.

Characters, Strings Data Types:

DATA TYPE	keyword	Description
Char	char	Maximum length of 8,000 characters(Fixed length non-Unicode characters)
Varchar	varchar	Maximum length of 8,000 characters(Variable length non-Unicode data)

Operator in SQL

- An operator is a reserved word or a character used primarily in an SQL statement's WHERE clause to perform operations(s), such as comparisons and arithmetic operations.
 - Arithmetic operators
 - Comparisons Operators
 - Logical operators
 - Operators used to negate the conditions